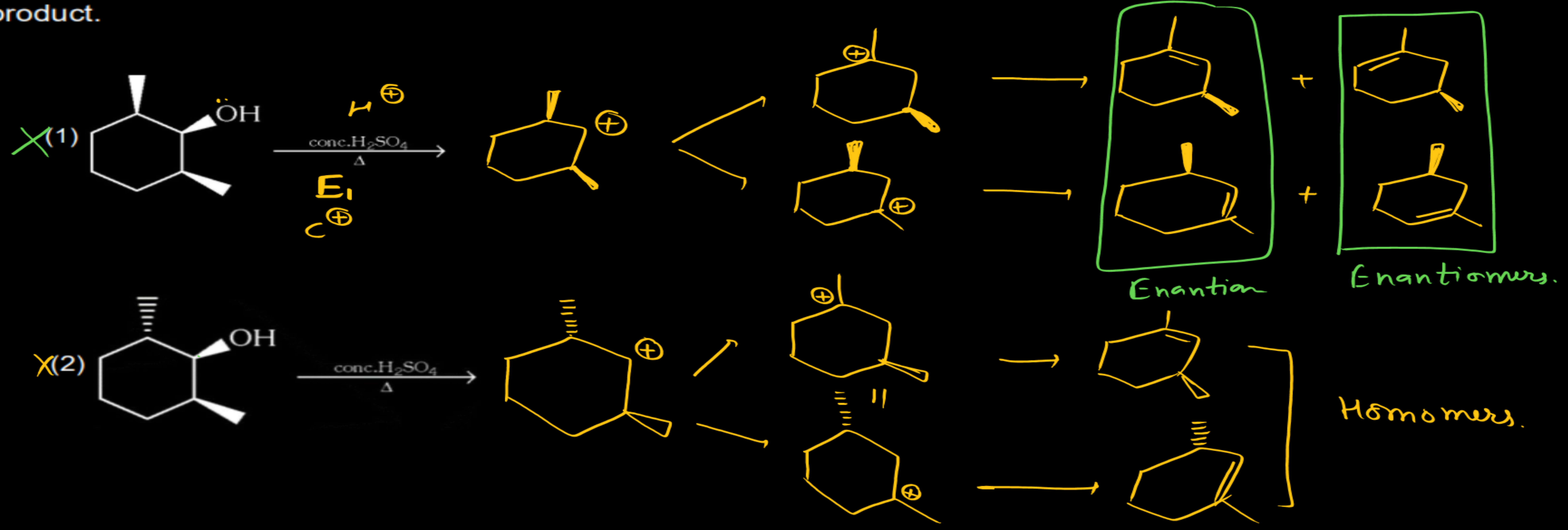
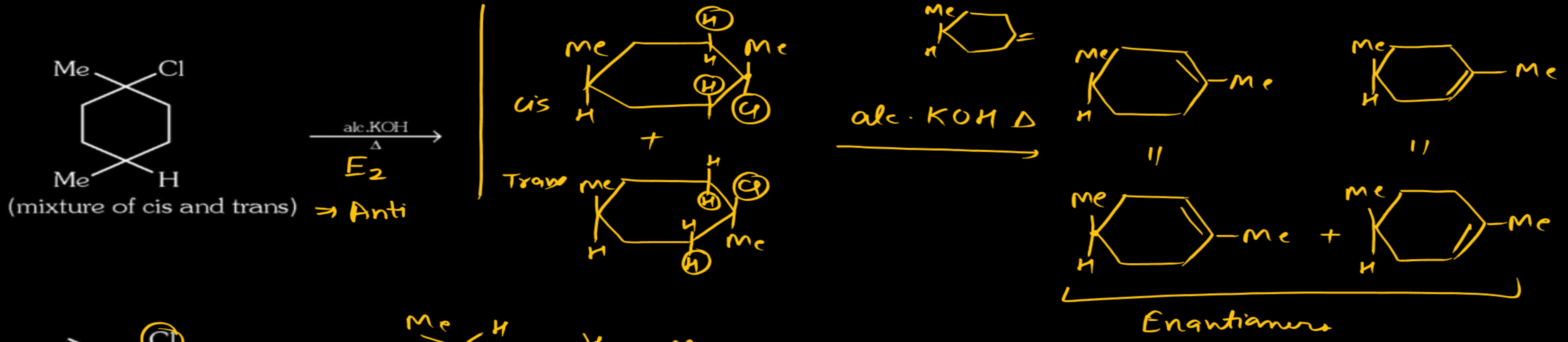
Competishun
Power of real gurus

CATS-PLUS
2022

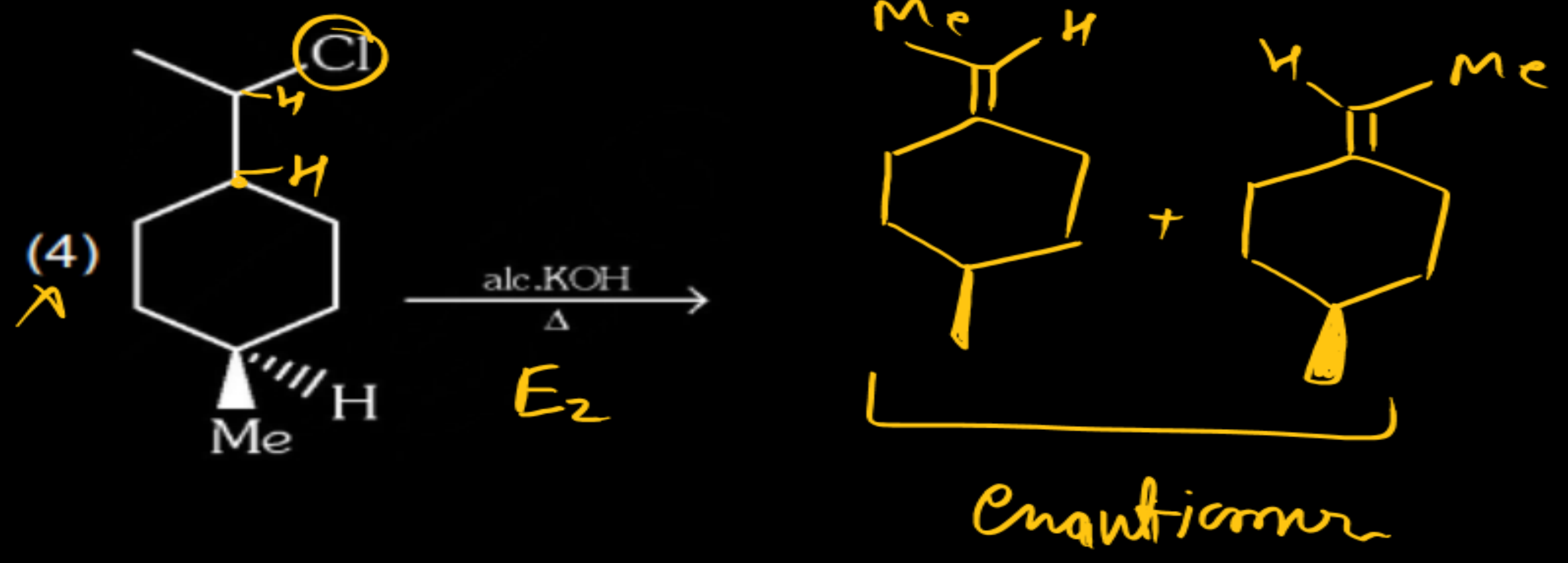
Give sum of serial number of the following reaction which give diastereomeric pair as major elimination product.



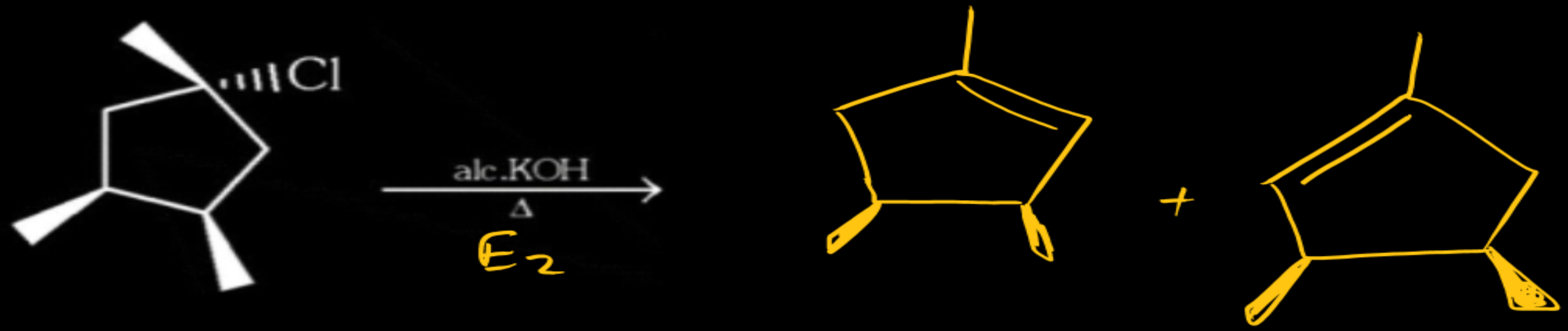
(3)


Me
Cl
Me
H
(mixture of cis and trans)
alc.KOH
Δ
E_2
⇒ Anti
cis
Trans
alc. KOH Δ
Enantiomers

(4)

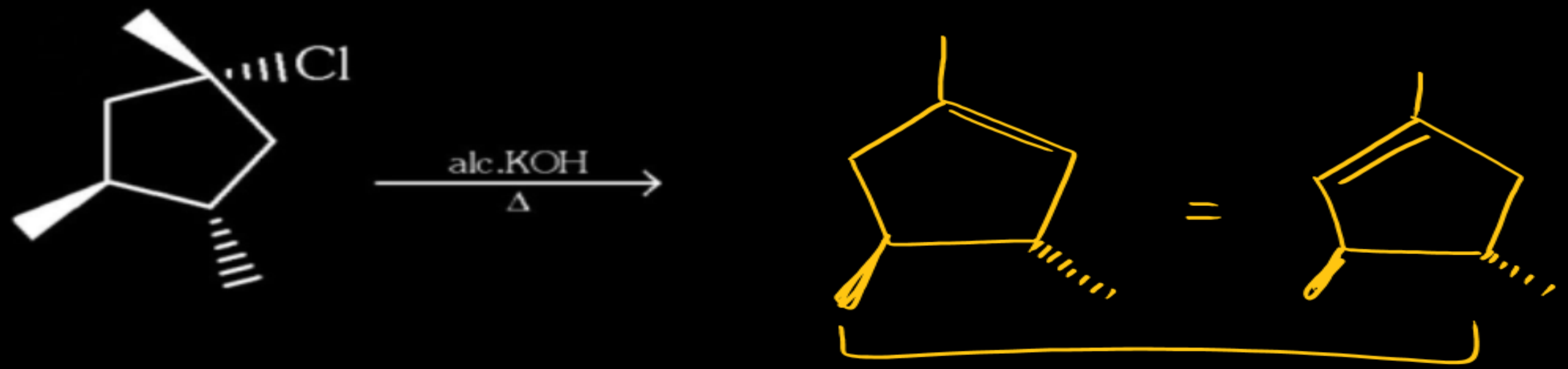

Cl
H
Me
alc.KOH
Δ
E_2
Me
H
H
Me
Enantiomer

(5)



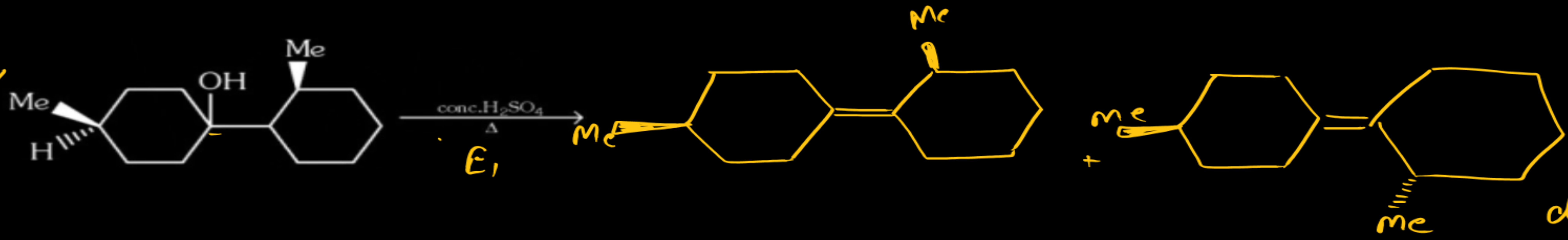
enatiomers.

(6)



Identical

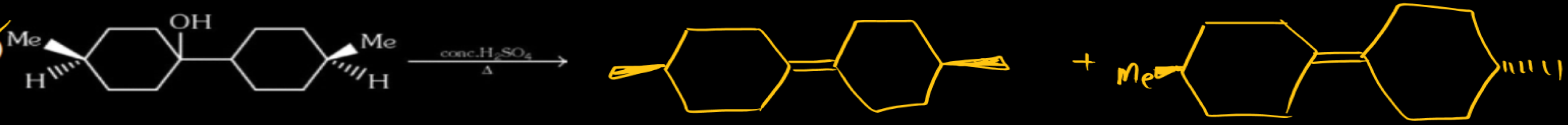
(7)



geo. Iso.

diastereomers

(8)



$\boxed{7+8=15}$

geo. Iso.

Q. Compound (X) (C_5H_8O) does not react appreciably with Lucas reagent at room temperature but gives a precipitate with ammonical silver nitrate. With excess of MeMgBr, 0.42 g of (X) gives 224 ml CH_4 at STP. Treatment of (X) with H_2 in presence of Pt catalyst followed by boiling with excess HI gives n–pentane. The correct statement(s) about 'X' is/are:

(A) compound 'X' ~~and~~ can give Baeyers reagent test

(B) Its monohydric alcohol.

(C) compound 'X' can be formed by pent-4ynal on reduction with LAH.

(D) Permole of 'X', gives 1 mole of H_2 on reaction with excess Na–metal.

M.wt: 60 + 8 + 16 = $\boxed{84}$

X (C_5H_8O)

DU = 2

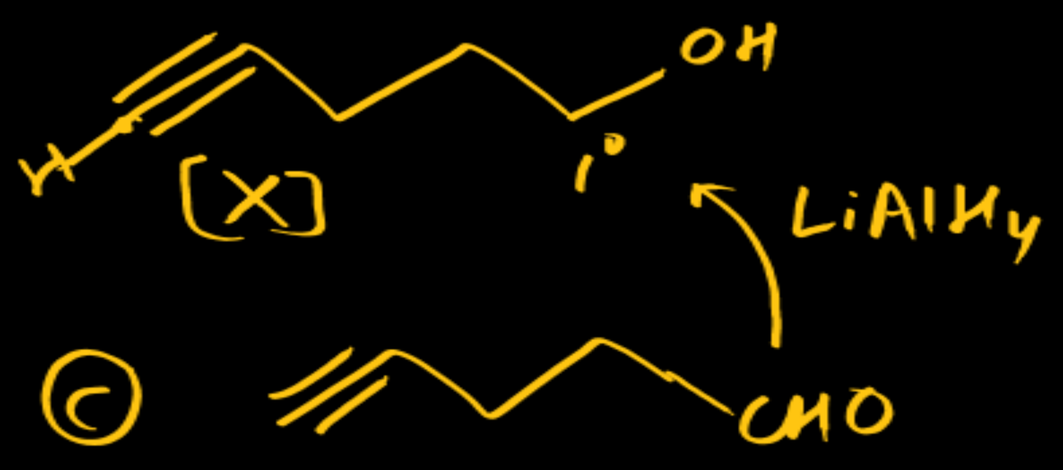
$\text{X} + CH_3MgBr \longrightarrow CH_4$

0.42 g ; 224 ml

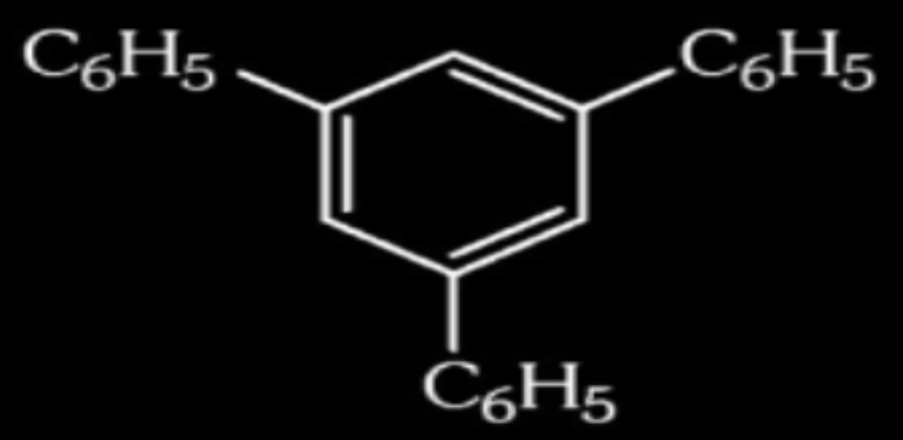
$\frac{0.42}{84} = \frac{1}{200}$; $\frac{224}{22400} = \frac{1}{100}$

2 acidic hydrogens.

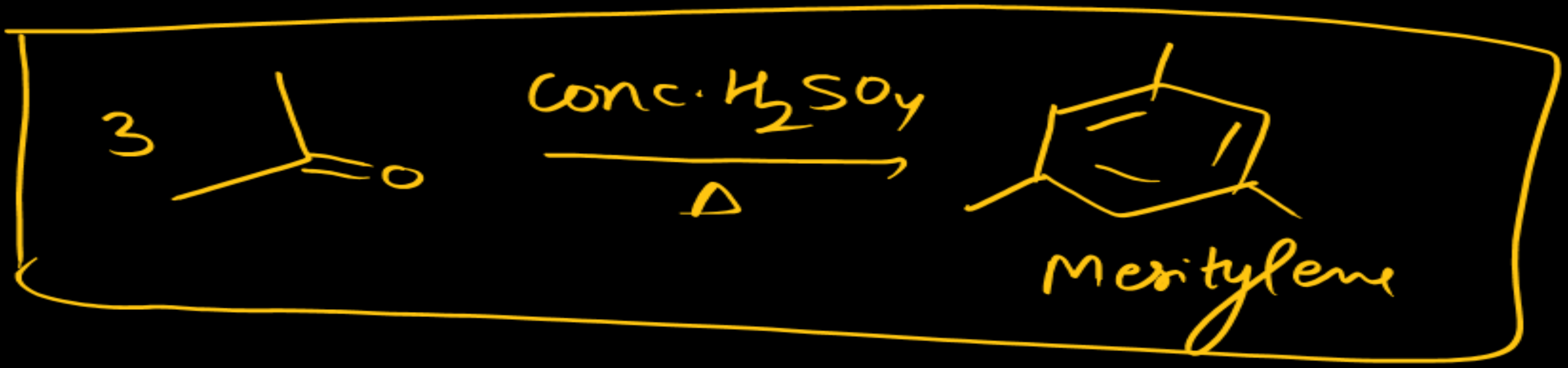
$H-C\equiv C-$ + 1° Alcohol

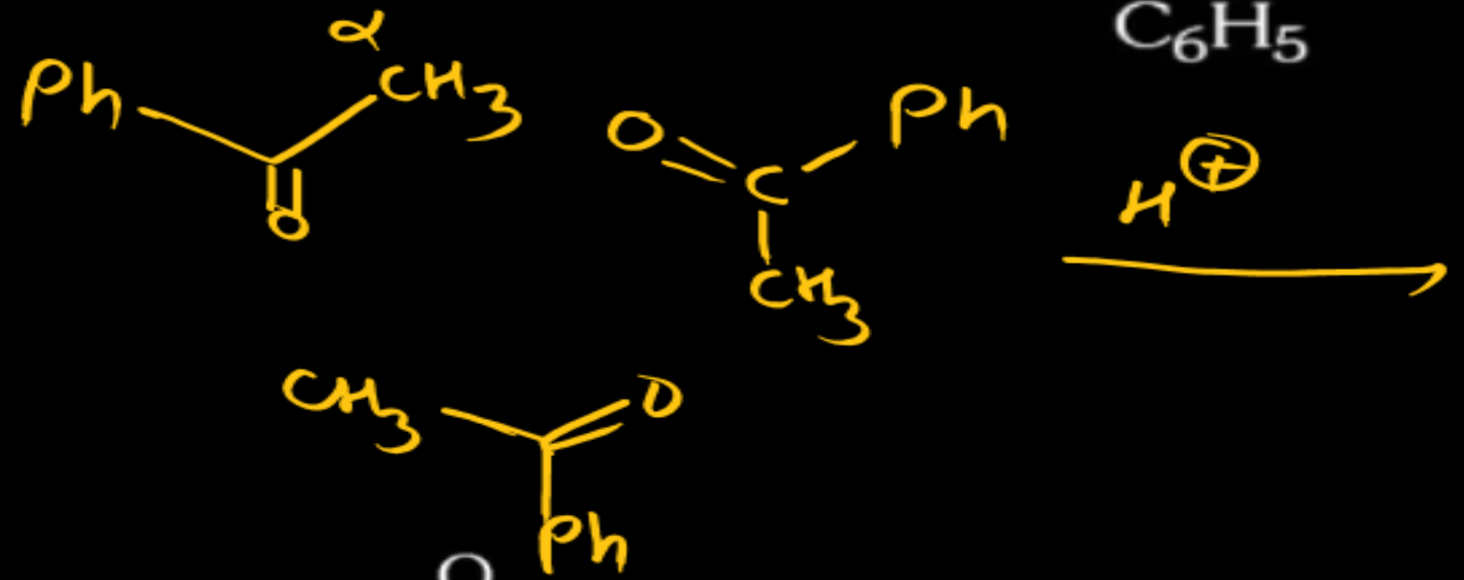


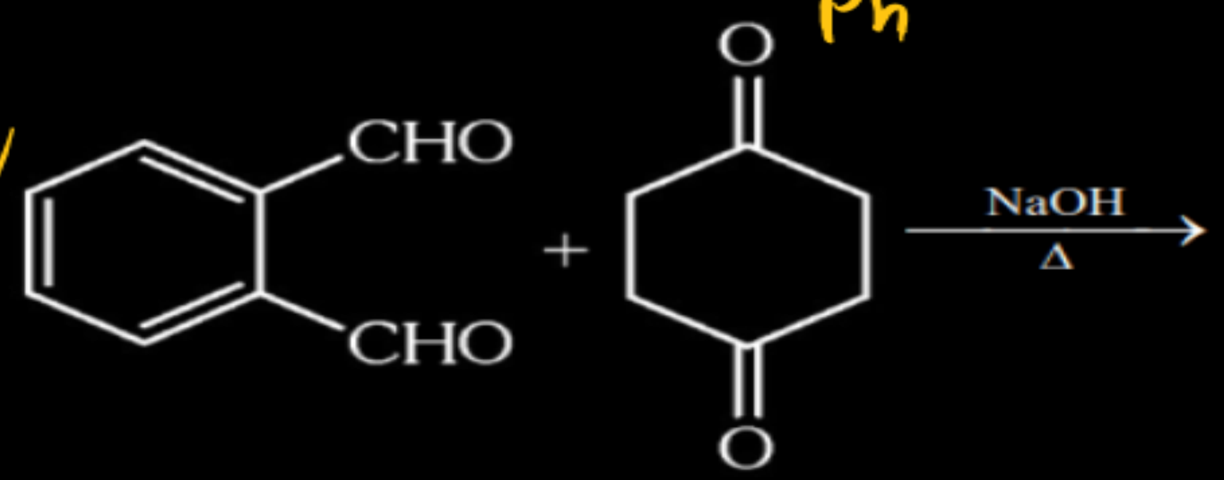
Q In which of the following reaction, reactants and products ars correctly matched.

(A) $C_6H_5-COCH_3 \xrightarrow{H^+}$ 

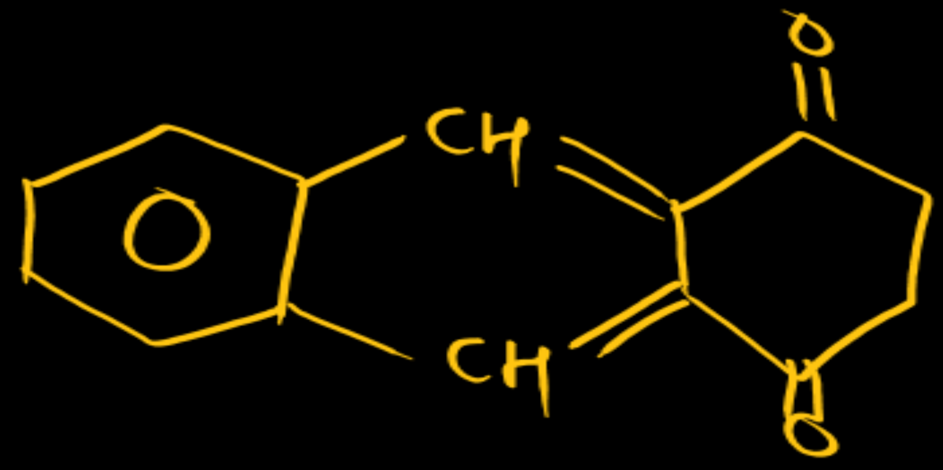
Acetone





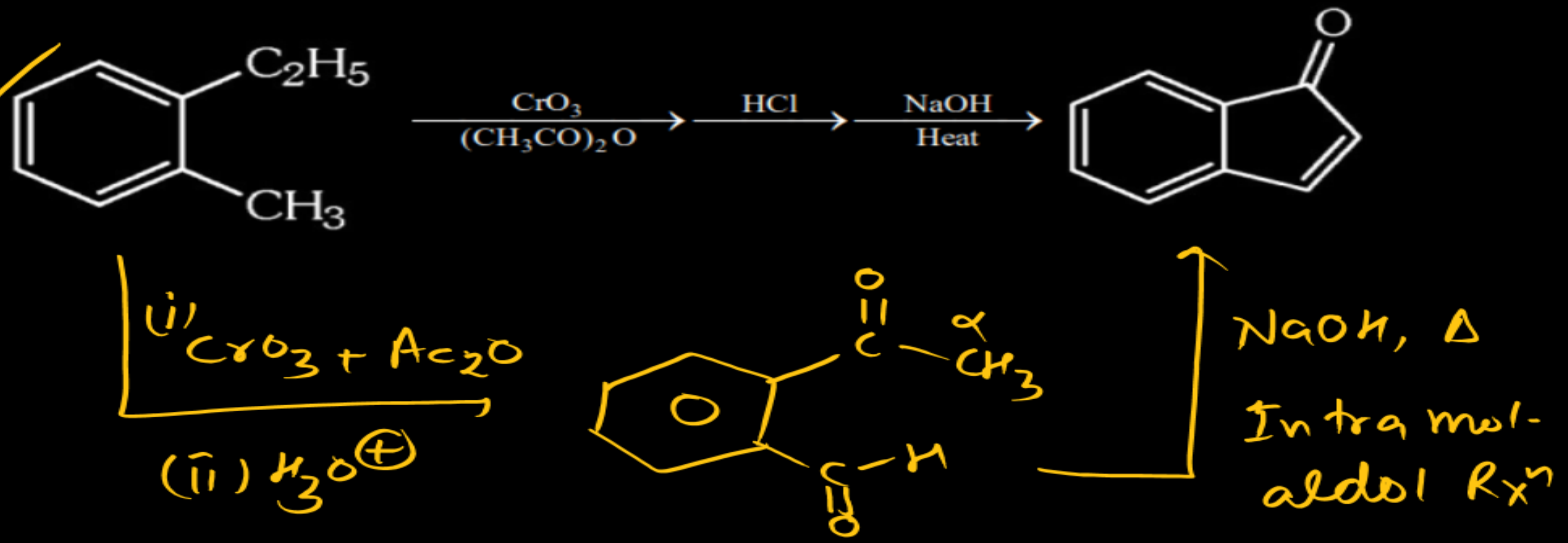
(B) 

Compound with DU 10



DU = 3 + 7 = 10

(C)



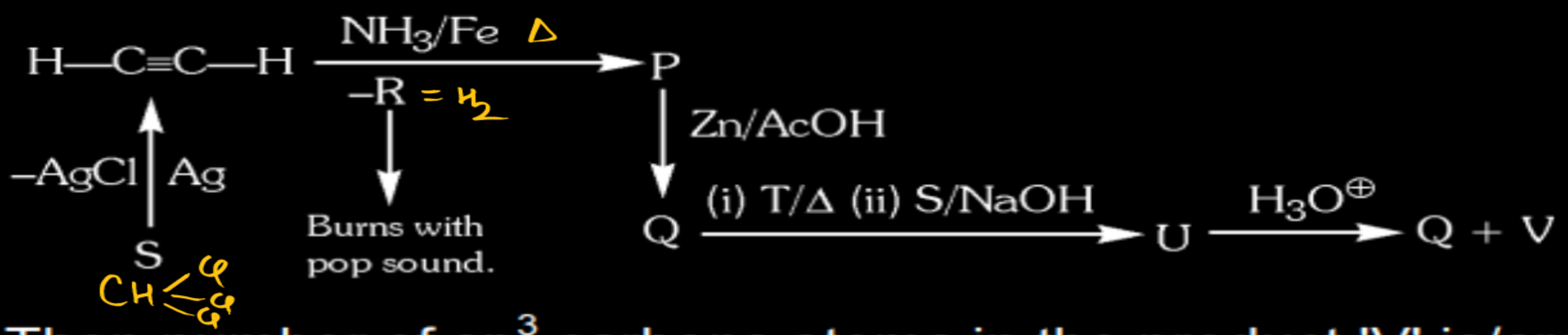
$$Ph-CH_3 \xrightarrow[(CH_3CO)_2O]{CrO_3} Ph-CHO$$

(D) $C_6H_5CHO + CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-CH_3 \xrightarrow[C_2H_5OH]{C_2H_5ONa \quad \Delta} (C_6H_5-CH=CH)_2C=O$

$$Ph-CH=O + \overset{\alpha}{CH_3}-\overset{\overset{\displaystyle O}{\|}}{C}-CH_3 + O=CH-Ph \longrightarrow Ph-CH=CH-\overset{\overset{\displaystyle O}{\|}}{C}-CH=CH-Ph$$

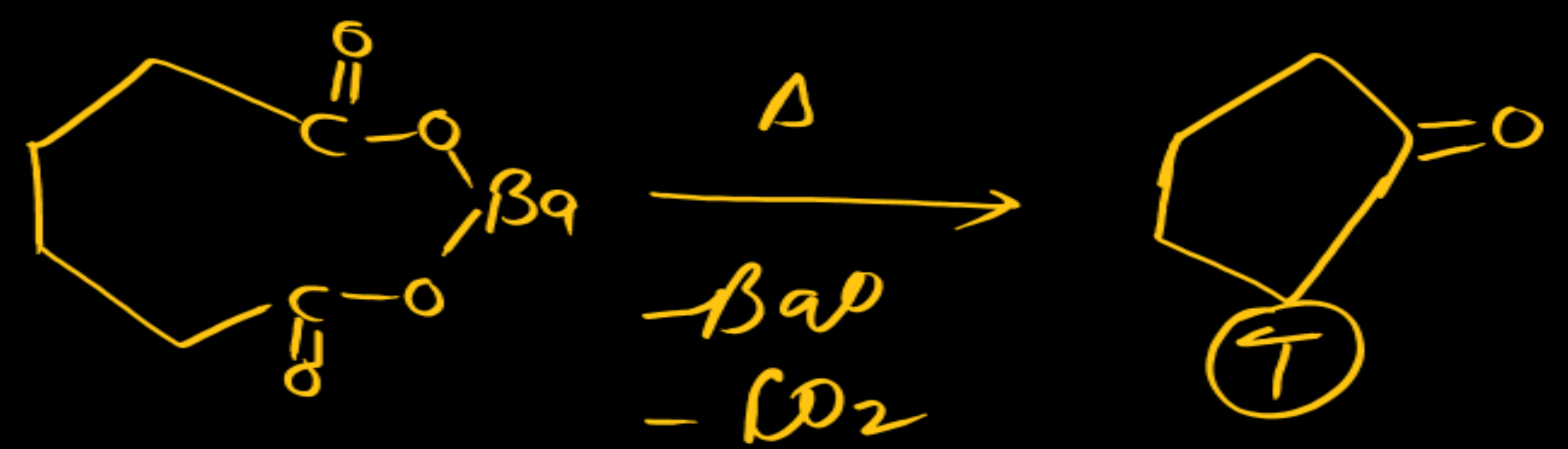
3 Product

Q. Barium adipate $\xrightarrow[-BaO,\ -CO_2]{\text{Drydistillation}}$ T

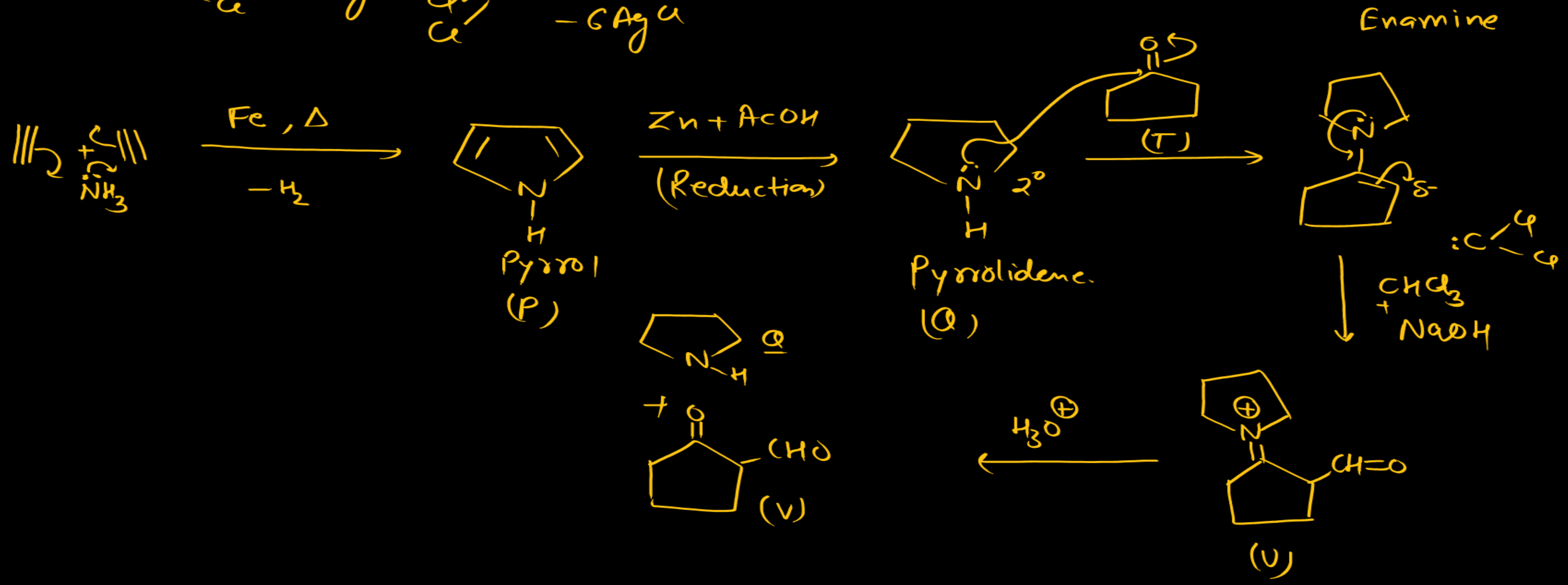


Then number of sp^3 carbons atoms in the product 'V' is/are:

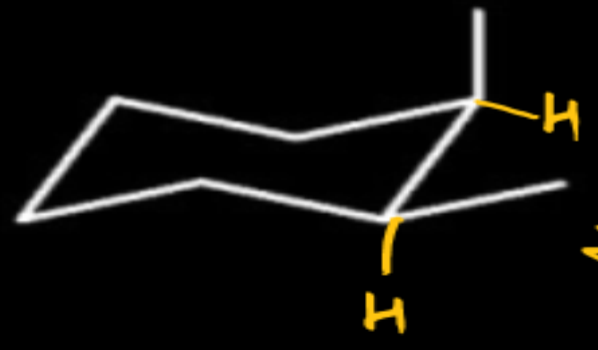
Sol



S $\quad CHCl_3 + 6Ag + CHCl_3 \xrightarrow[-6AgCl]{\Delta} H-C\equiv C-H$



Q Select incorrect statement.

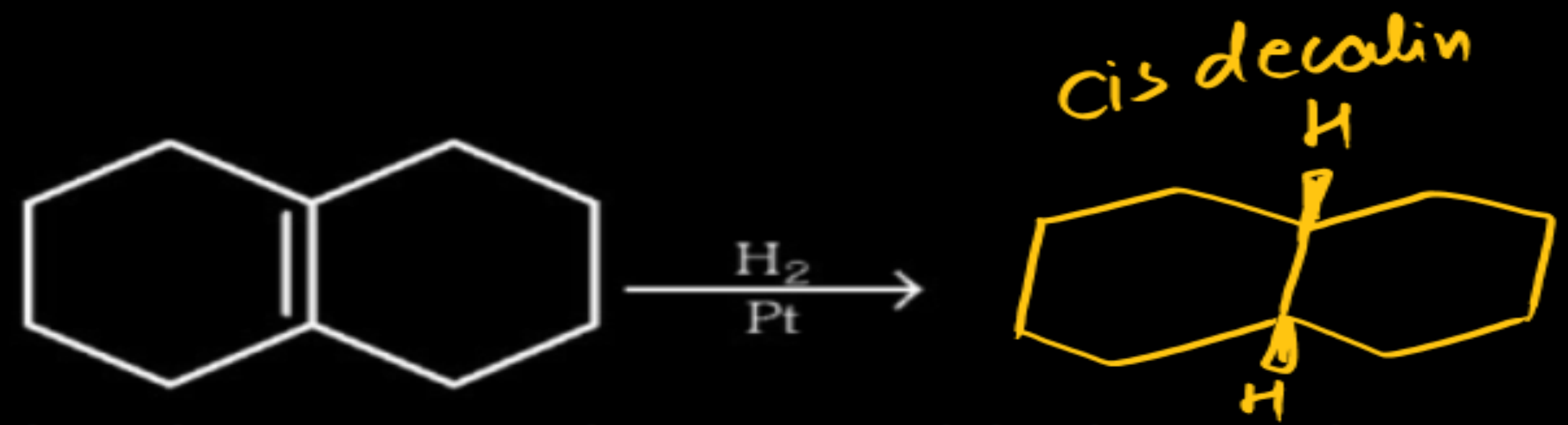
(A)  and  are non super imposable mirror images of each other.

(Conformational enantiomers)

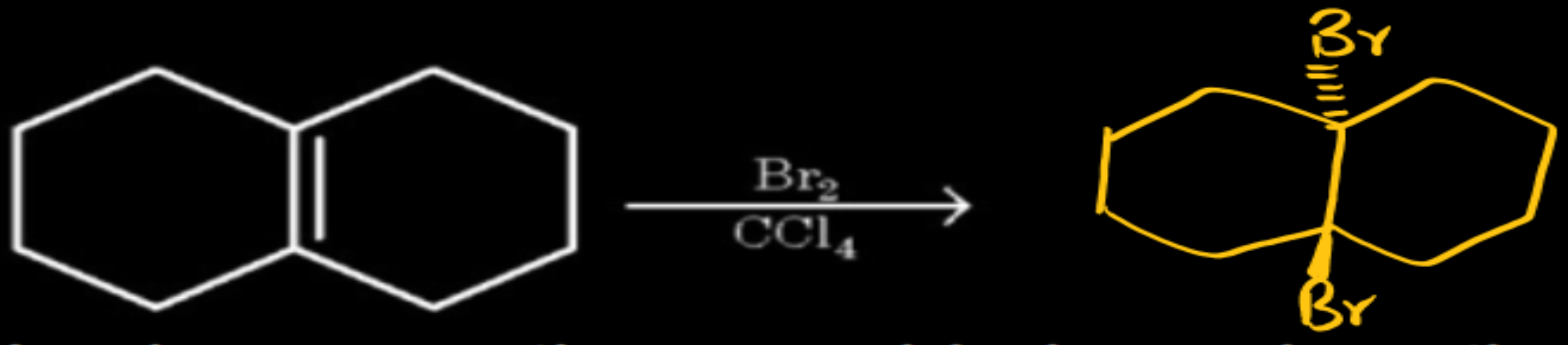
cis

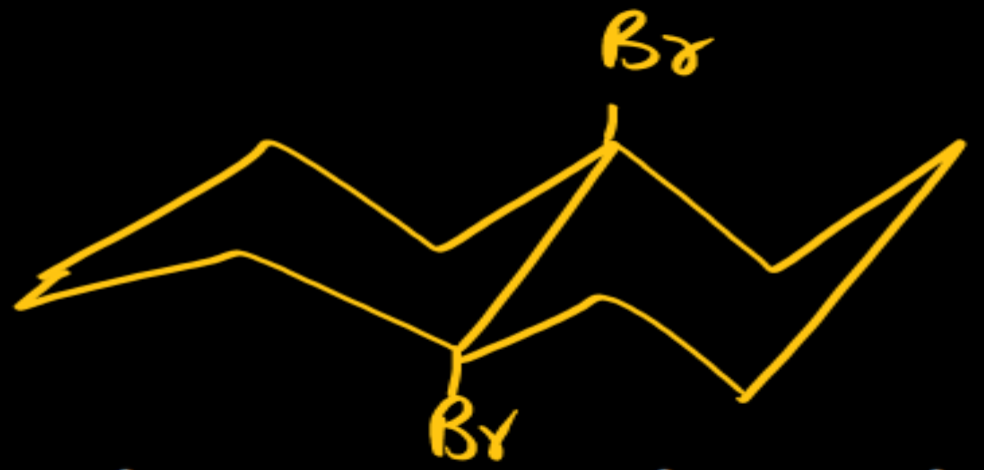
1,2-Dimethyl cyclohexane

(B)  and  are configurational isomers.

(C) 

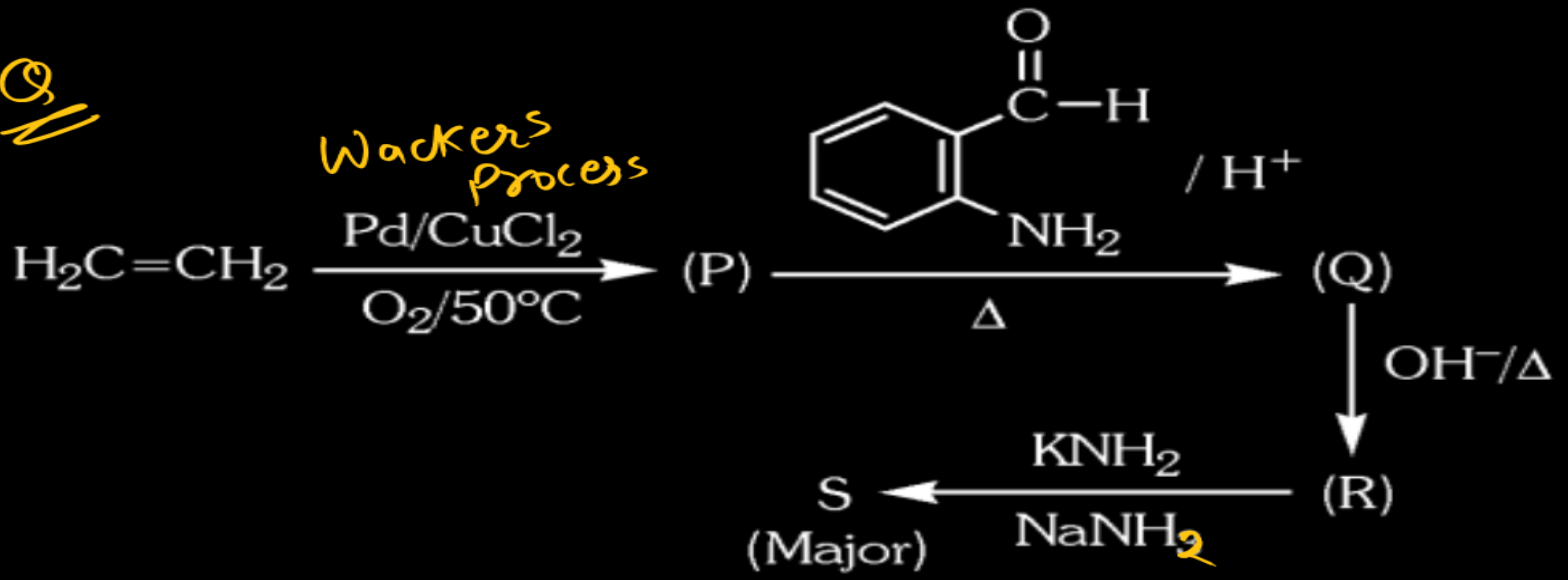
In above reaction, considering only syn addition, pair of non resolvable organic products are obtained.

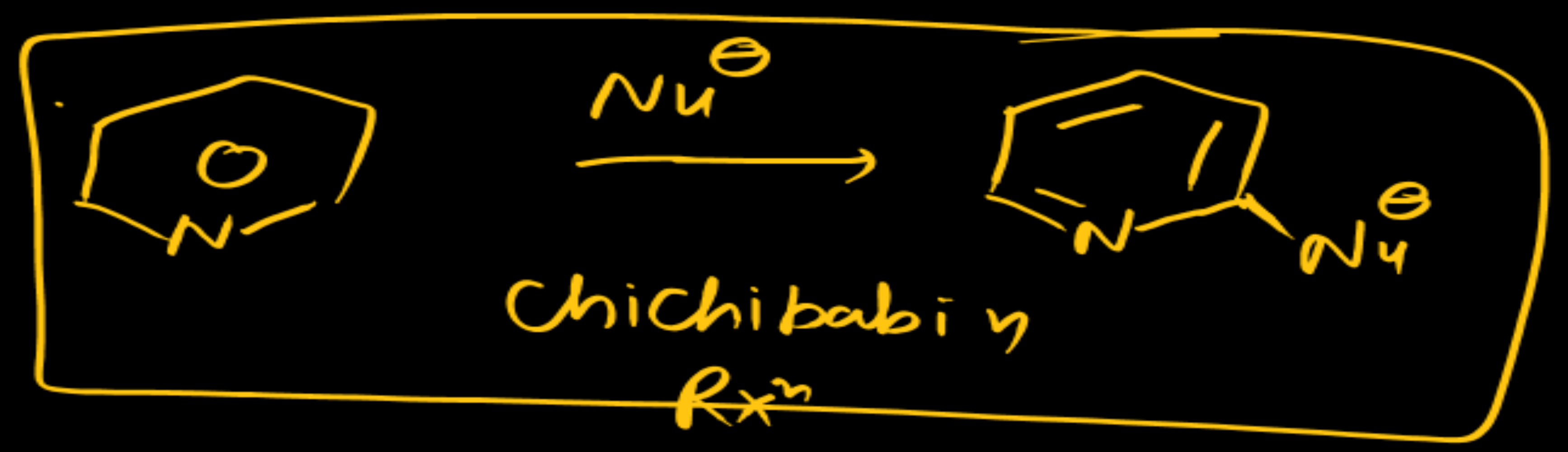
(D) 



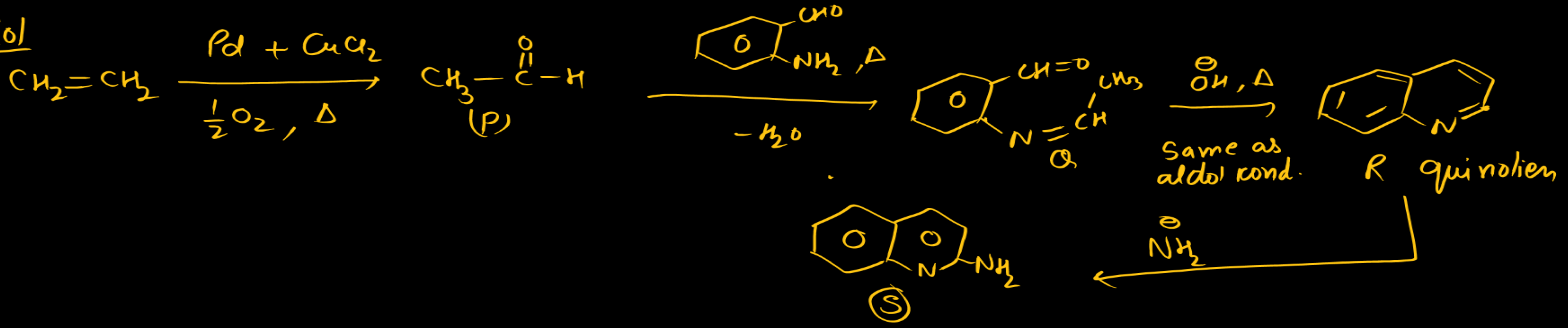
In above reaction, considering only anti addition only one organic product is formed.

Q


Wackers process
$H_2C=CH_2$
$Pd/CuCl_2$
$O_2/50°C$
(P)
O
C—H
NH_2
/ H^+
Δ
(Q)
OH^-/Δ
(R)
KNH_2
$NaNH_2$
S
(Major)

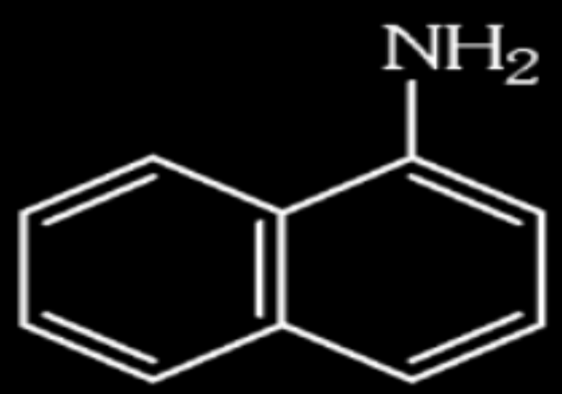

$Nu^{\ominus}$
N
$Nu^{\ominus}$
Chichibabin Rxn

Sol


$CH_2=CH_2$
Pd + $CuCl_2$
$\frac{1}{2}O_2$, Δ
$CH_3-\overset{O}{\overset{||}{C}}-H$
(P)
CHO
NH_2 , Δ
$-H_2O$
CH=O
CH_3
N=CH
Q
$\overset{\ominus}{O}H$, Δ
Same as aldol cond.
N
R quinolien
$\overset{\ominus}{N}H_2$
N
NH_2
S

(A) $P = \underset{|}{\underset{OH}{CH_2}}-\underset{|}{\underset{OH}{CH_2}}$

(B) R = 2° amine

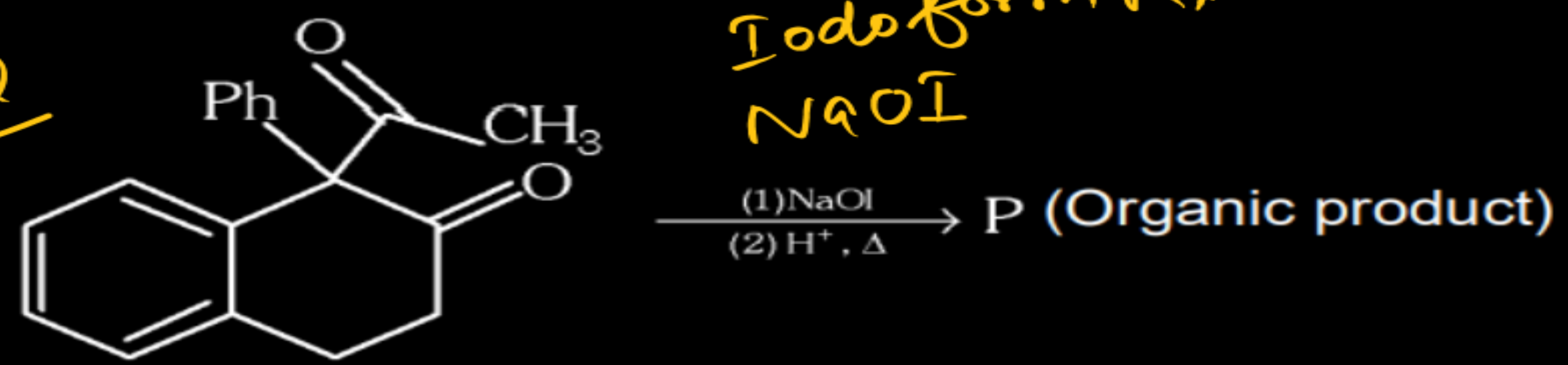
(C) S = 

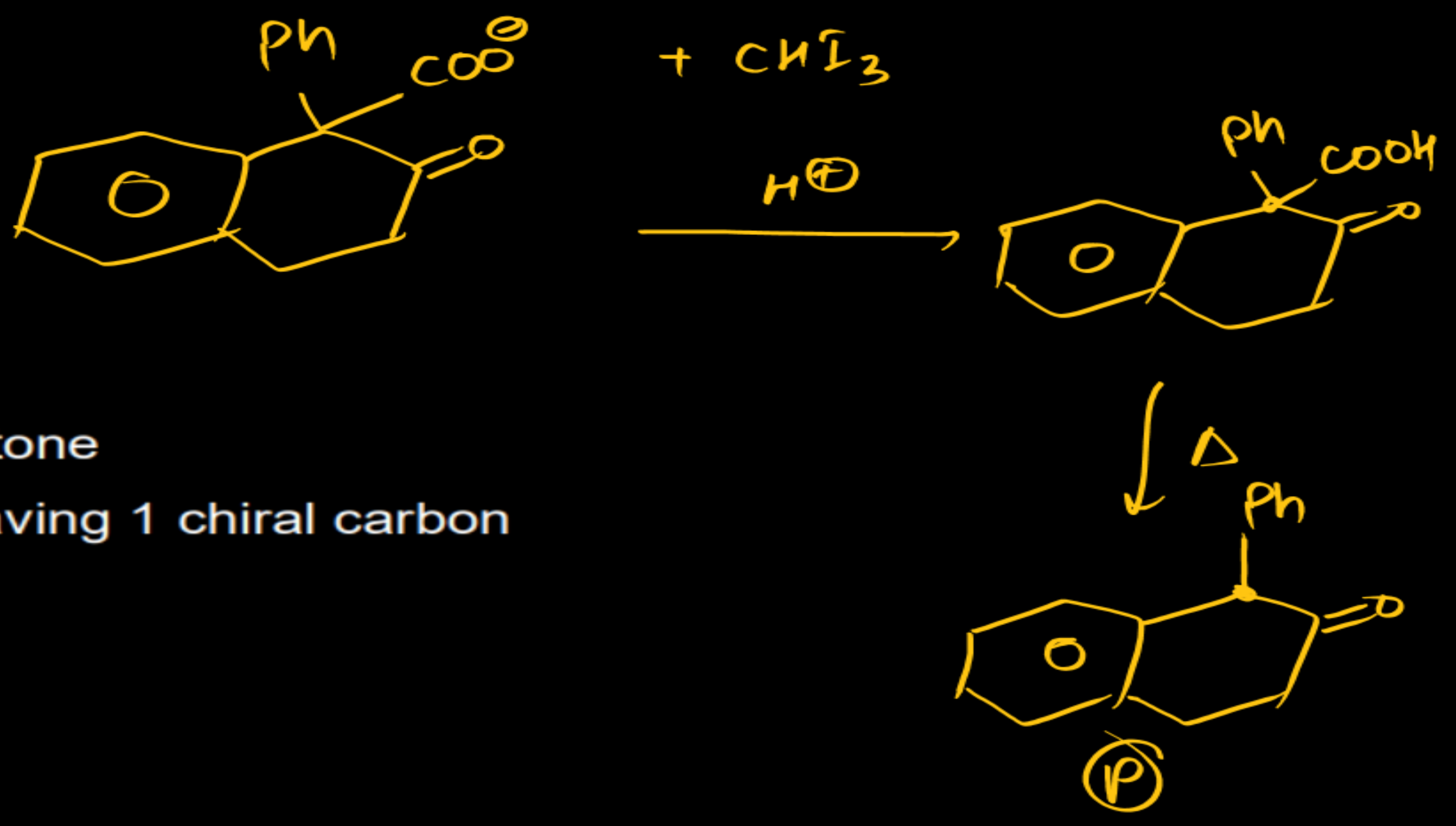
(D) R = 3° amine

Q

Iodoform Rxn

NaOI





The correct statement(s) about 'P' is/are:

(A) 'P' is carboxylic acid

(B) 'P is ketone

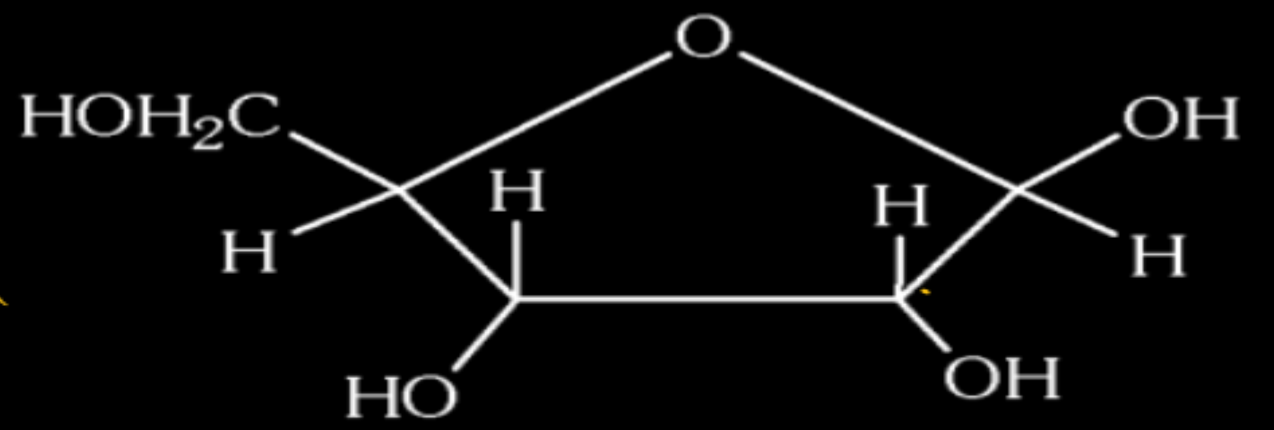
(C) 'P' is aldehyde

(D) 'P' is having 1 chiral carbon

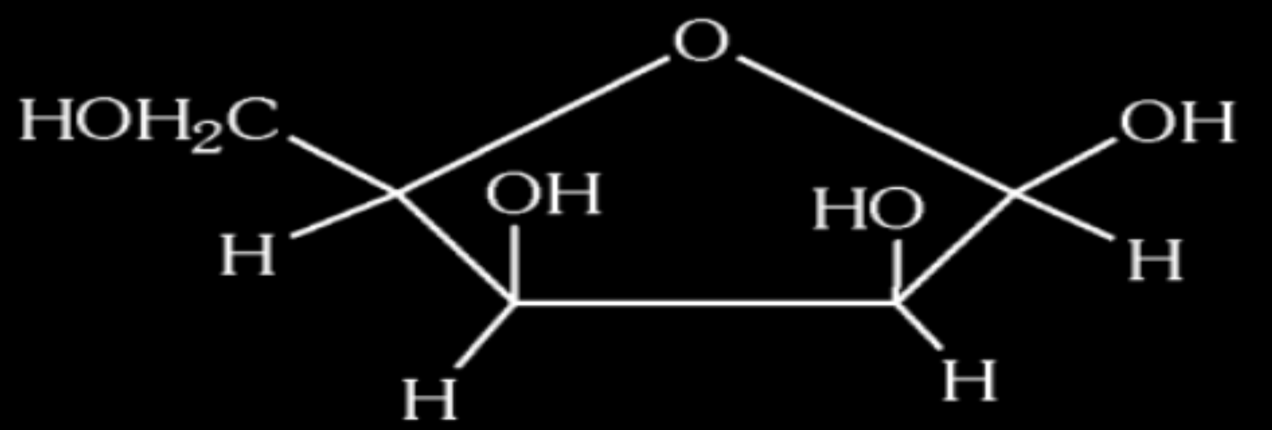
Q. All the OH group of β–pyranose form of D–xylose are equatorial. Which of the following is β–furanose of D–xylose.

Pentose

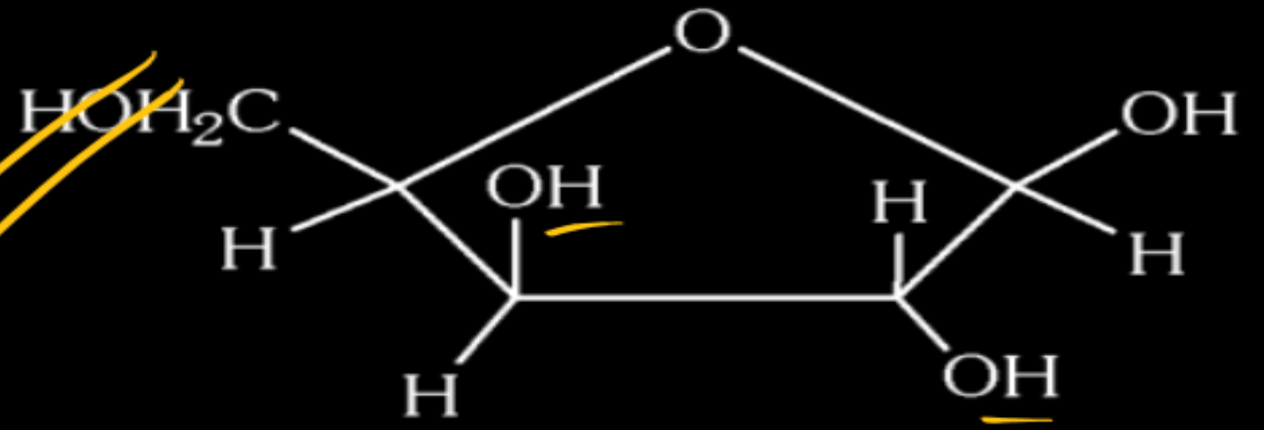
(A)



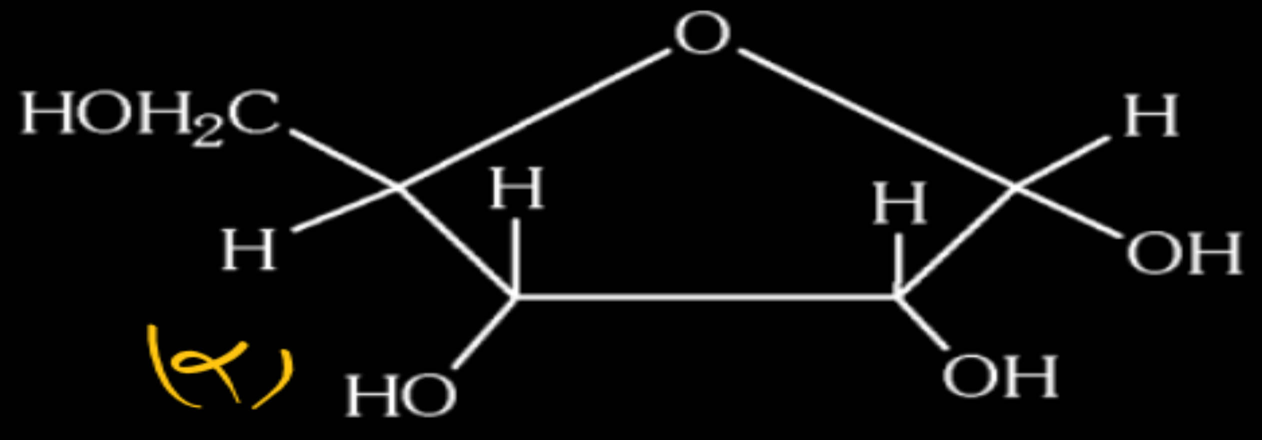
(B)



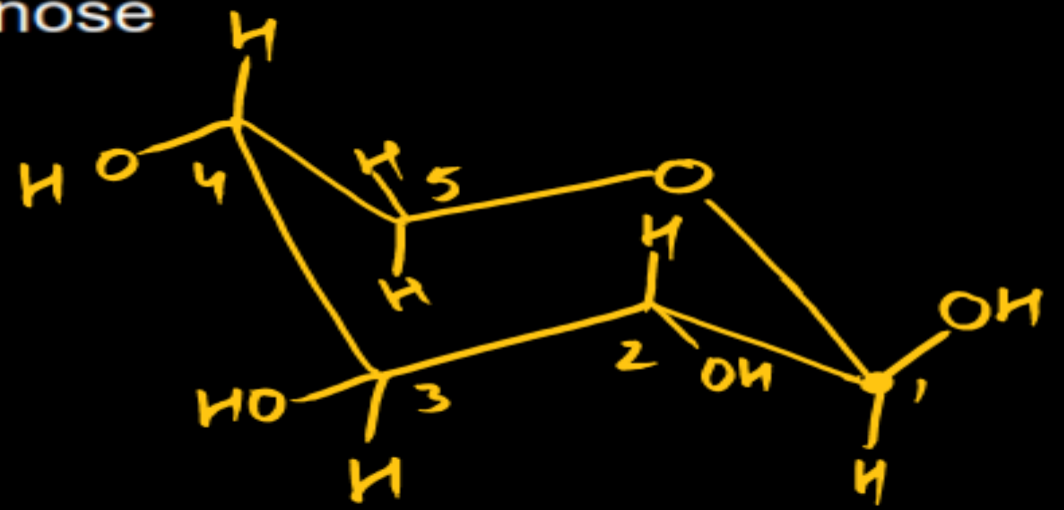
(C)



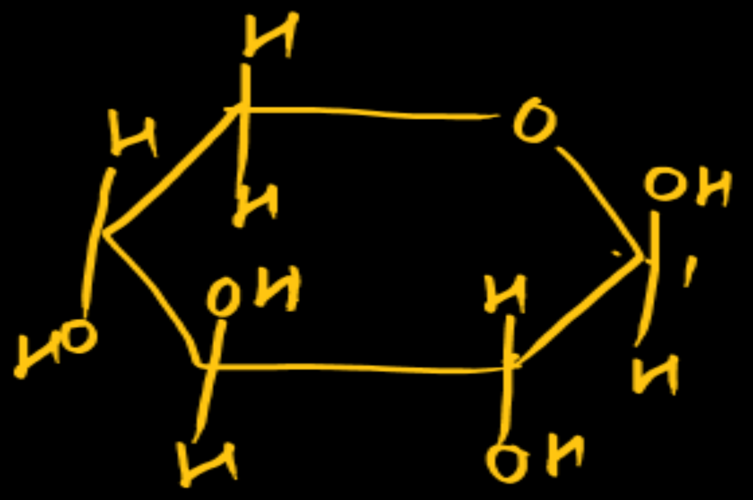
(D)

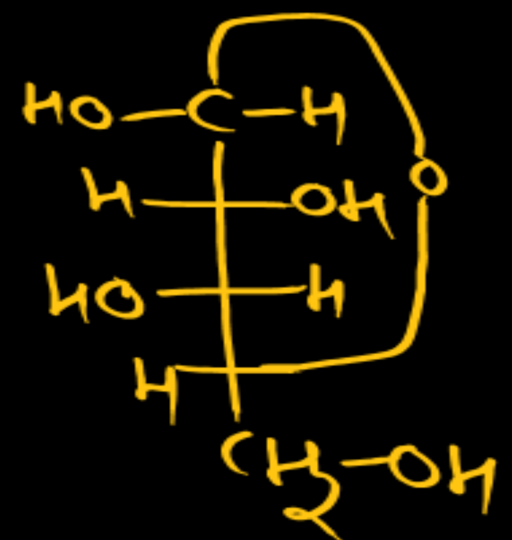


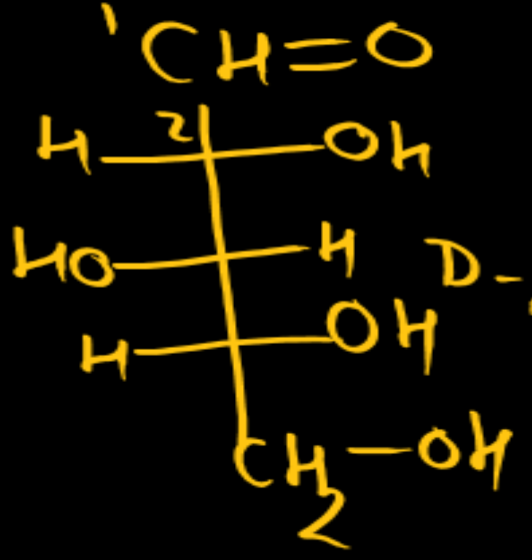
(α)



D-xylopyranose

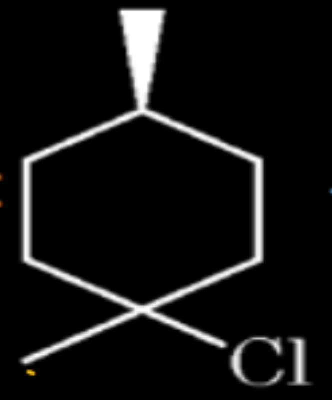


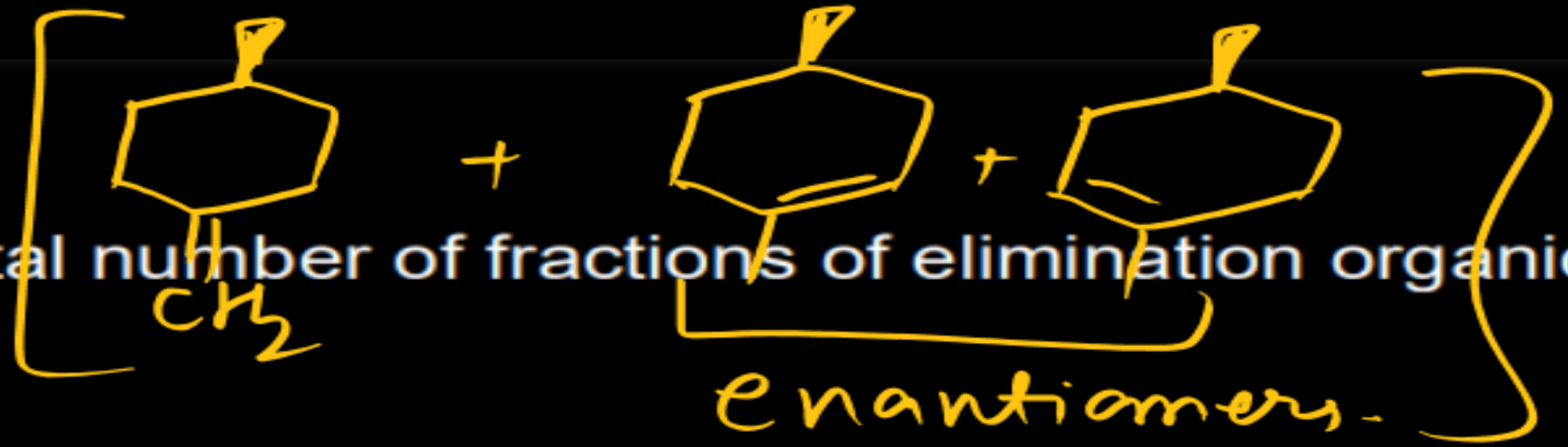




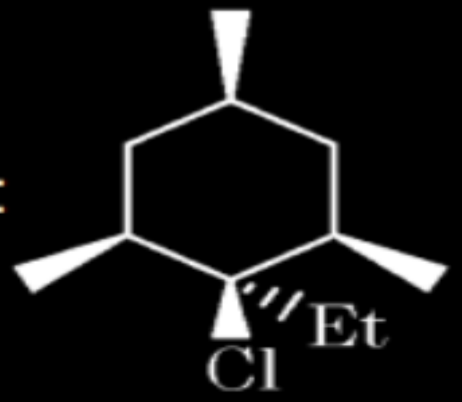
D-Xylose

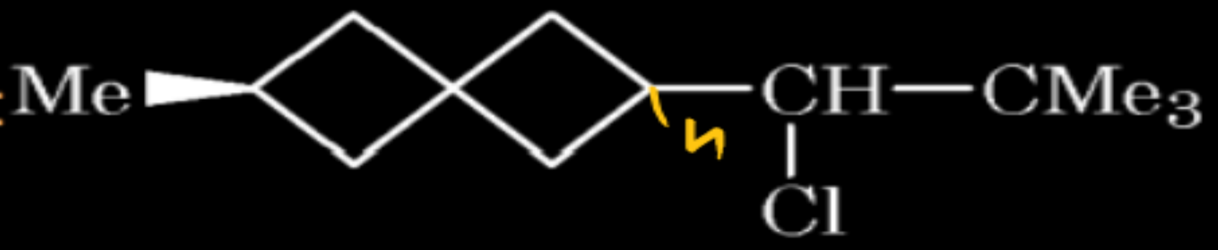
Q3

Reaction -I :  $\xrightarrow{\text{alc.KOH}}$ Total number of fractions of elimination organic products (P)



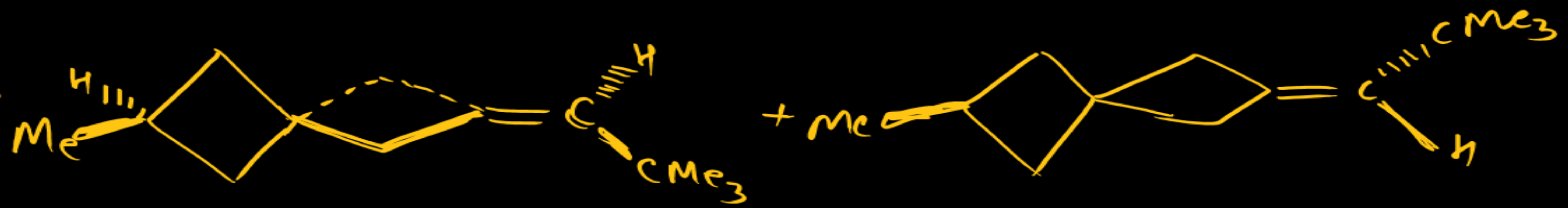
2

Reaction -II :  $\xrightarrow{\text{alc.KOH}}$ Total number of fractions of elimination organic products (Q)

Reaction -III :  $\xrightarrow{\text{alc.KOH}}$ Total number of fractions of elimination organic products (R)

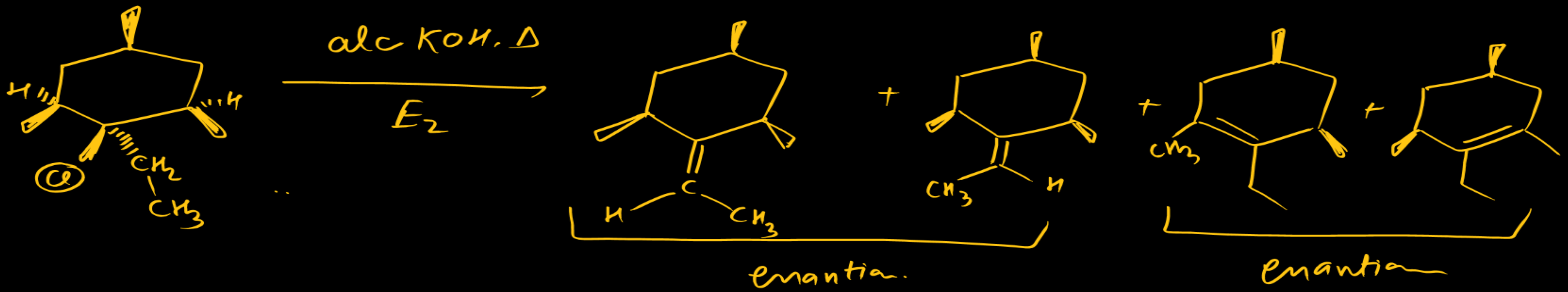
Give your answer in 3 digits as PQR.

(2, 2, 2)



geo. Iso. (2)

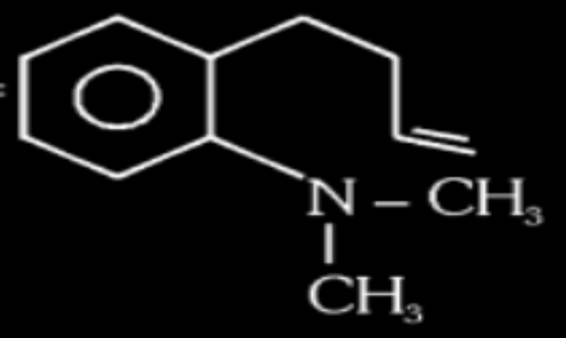
II

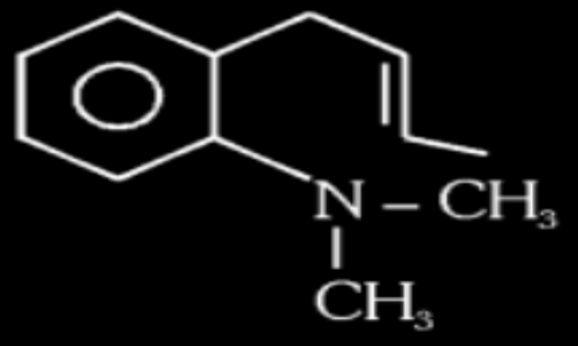


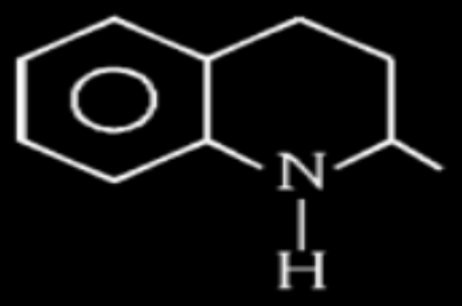
2

Compound 'M' $C_{10}H_{13}NO$ is soluble in dil. HCl. On oxidation with $KMnO_4$, compound (M) gives anthranillic acid. On treatment with $NaNO_2$/conc. HCl at 0° followed by treatment with H_3PO_2/H_2O compound (M) gives compound (N) $C_{10}H_{12}O$. Compound (M) as well as N give haloform test. In the given reaction sequence Q and R are:

$$M \xrightarrow[PH=5]{H^+,\Delta} P \xrightarrow{Ni/H_2} Q \xrightarrow[(ii)\,AgOH/\Delta]{(i)\,MeI\,(excess)} R$$

(A) R = 

(B) R = 

(C) Q = 

(D) Q =